

Pharmacy Management System Database Project

Project Title

Pharmacy Management System Using SQL Server

Project Description

The Pharmacy Management System is a database project designed to help pharmacies manage medicines, customers, suppliers, sales, payments, complaints, and medicine requests efficiently.

The system aims to:

- Organize pharmacy operations
- Store medicine information
- Manage sales and supplier invoices
- Track stock quantities
- Handle customer requests and complaints
- Improve data accuracy
- Reduce manual work

The project was developed using:

- **SQL Server**
 - **ERD (Entity Relationship Diagram)**
 - **Relational Database Design**
 - **SQL Queries**
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Project Objectives

Main Objectives

- Improve pharmacy management operations
 - Reduce human errors
 - Facilitate data storage and retrieval
 - Support decision-making using organized data
 - Improve customer service
 - Track medicine stock levels
 - Manage supplier relationships
 - Store customer complaints and requests
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Technologies Used

Technology	Purpose
SQL Server	Database Management
ERDPlus	ERD Design
SQL	Database Queries
GitHub	Project Hosting

System Features

- Medicine Management
- Customer Management
- Supplier Management
- Sales Management
- Invoice Tracking
- Payment Management

- Complaint Tracking
 - Medicine Request System
 - Category Management
 - Inventory Monitoring
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Database Entities

1. Categories

Stores medicine categories.

Attributes

- CategoryID (PK)
 - CategoryName
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2. Medicines

Stores medicine information.

Attributes

- MedicineID (PK)
 - Name
 - Description
 - Price
 - QuantityInStock
 - MinimumStockLevel
 - CategoryID (FK)
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3. Customers

Stores customer information.

Attributes

- CustomerID (PK)
 - Name
 - Phone
 - Address
-

4. Sales

Stores sales transactions.

Attributes

- SalesID (PK)
 - SalesDate
 - TotalAmount
 - CustomerID (FK)
-

5. SalesDetails

Stores medicines included in sales.

Attributes

- SaleDetailID (PK)
 - SaleID (FK)
 - MedicineID (FK)
 - Quantity
 - Price
-

6. Suppliers

Stores supplier information.

Attributes

- SupplierID (PK)
 - Name
 - Phone
 - Address
-

7. SupplierInvoices

Stores supplier invoices.

Attributes

- InvoiceID (PK)
 - InvoiceDate
 - TotalAmount
 - SupplierID (FK)
-

8. InvoiceDetails

Stores medicines inside supplier invoices.

Attributes

- InvoiceDetailID (PK)
 - InvoiceID (FK)
 - MedicineID (FK)
 - Quantity
-

9. Payment

Stores payment information.

Attributes

- PaymentID (PK)

- Amount
- Method
- PurchaseID

10. Request

Stores customer medicine requests.

Attributes

- RequestID (PK)
- CustomerID (FK)
- MedicineName
- RequestDate
- Status

11. Complaint

Stores customer complaints.

Attributes

- ComplaintID (PK)
- CustomerID (FK)
- ComplaintDate
- Description
- Status

Relationships

Relationship	Description
Categories → Medicines	One category contains many medicines
Customers → Sales	One customer can make many sales

Relationship	Description
Sales → SalesDetails	One sale contains many sale details
Medicines → SalesDetails	One medicine can appear in many sales
Suppliers → SupplierInvoices	One supplier has many invoices
SupplierInvoices → InvoiceDetails	One invoice contains many medicines
Medicines → InvoiceDetails	One medicine appears in many invoices
Customers → Complaints	One customer can make many complaints
Customers → Requests	One customer can make many requests
Sales → Payment	One sale has one payment

Business Rules

1. Each medicine belongs to one category.
 2. A customer can make multiple purchases.
 3. Each sale must belong to one customer.
 4. Medicines can appear in multiple sales.
 5. Suppliers provide medicines through invoices.
 6. Payments are linked to sales.
 7. Customers can submit complaints.
 8. Customers can request unavailable medicines.
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Normalization

The database design follows normalization principles:

First Normal Form (1NF)

- No repeating groups
- Atomic values

Second Normal Form (2NF)

- All non-key attributes depend on the primary key

Third Normal Form (3NF)

- No transitive dependency
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Advantages of the System

- Easy data management
 - Fast searching and reporting
 - Better inventory control
 - Improved customer service
 - Reduced paperwork
 - Organized supplier tracking
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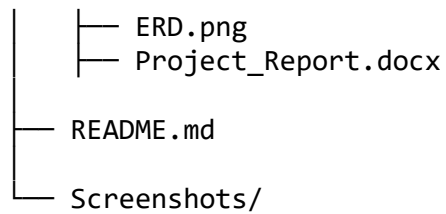
Future Improvements

- Add employee management
 - Add authentication system
 - Add barcode scanning
 - Add online ordering
 - Add reporting dashboard
 - Add expiration date tracking
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GitHub Repository Structure

Pharmacy-Management-System/

```
|
|— Database/
|   |— create_database.sql
|   |— insert_data.sql
|   |— queries.sql
|— Documentation/
```



README.md Content for GitHub

Pharmacy Management System

A SQL Server database project for managing pharmacy operations including medicines, customers, suppliers, sales, payments, complaints, and requests.

Features

- Medicine management
- Sales tracking
- Customer management
- Supplier management
- Inventory management

Technologies

- SQL Server
- SQL
- ERDPlus

Database Modules

- Medicines
- Categories
- Sales
- Customers
- Suppliers
- Payments

- Complaints
- Requests

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